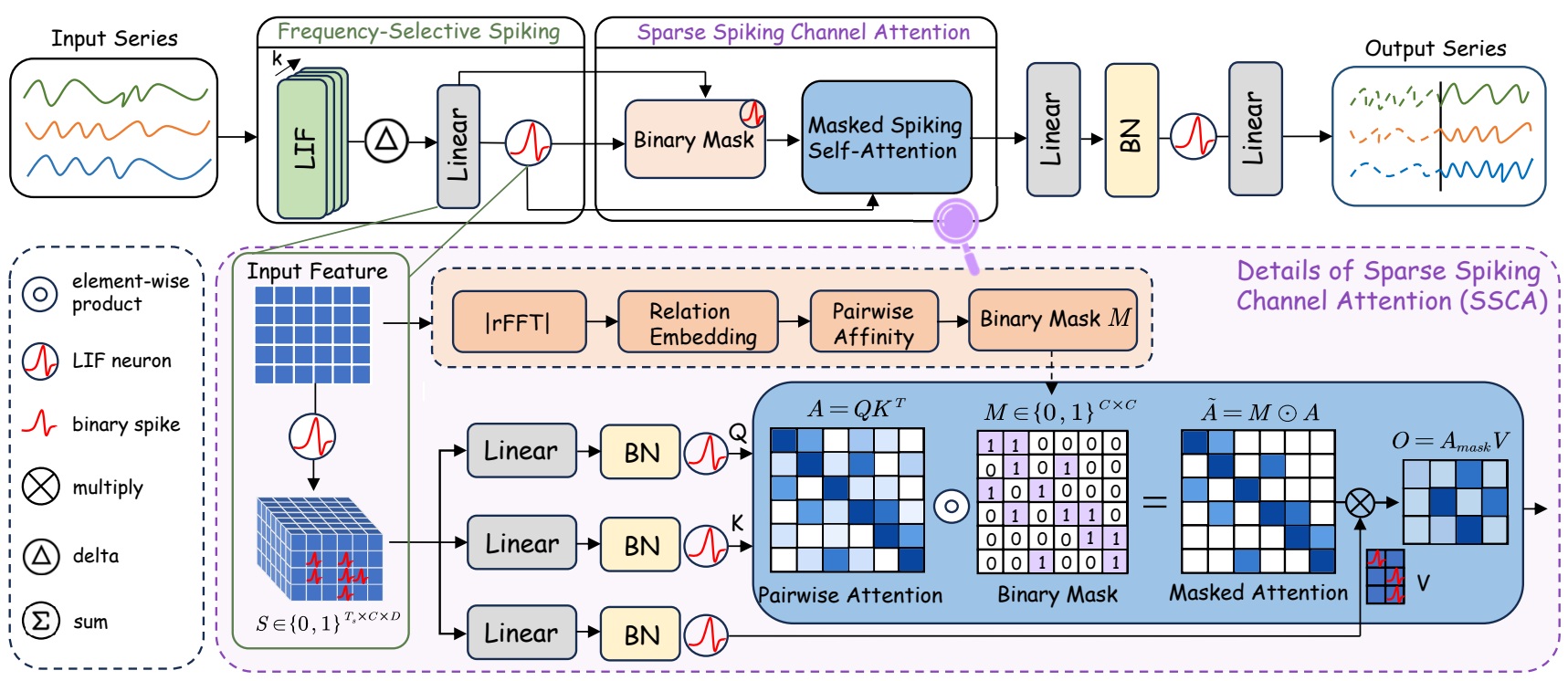

Input Series
Frequency-Selective Spiking
k
LIF
Linear
Sparse Spiking Channel Attention
Binary Mask
Masked Spiking Self-Attention
Linear
BN
Linear
Output Series
element-wise product
LIF neuron
binary spike
multiply
delta
sum
Input Feature
$S \in \{0,1\}^{T_s \times C \times D}$
|rFFT|
Relation Embedding
Pairwise Affinity
Binary Mask M
Details of Sparse Spiking Channel Attention (SSCA)
Linear
BN
Q
Linear
BN
K
Linear
BN
$A = QK^T$
$M \in \{0,1\}^{C \times C}$
$\tilde{A} = M \odot A$
$O = A_{mask}V$
Pairwise Attention
Binary Mask
Masked Attention
V